



Highly regioselective S_N2' reaction of β -fluoroallylic phosphates with organocopper reagents and its application to the synthesis of fluorine-containing carbocycles

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ABSTRACT

Treatment of β -fluoroallyl phosphates with 2.2 equiv. of organocuprates, prepared readily from 2.2 equiv. each of CuCN and organometallics (organolithium reagents, Grignard reagents, and organozinc reagents), in THF at -40 to 0 °C for 0.25 to 24 h gave γ -products in a highly regioselective manner. Thus obtained γ -adducts were subjected to the ring-closing metathesis, the corresponding cyclic fluoroalkenes being afforded in good yields.

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1. Introduction

Allylic substitution reactions with organocopper reagents, shown in Scheme 1, have been extensively investigated over the last few decades. In consequence, the process has become reasonably understood in the mechanistic aspect, and the control of the chemo-, the regio- and the stereo-selectivity for the reaction has been made possible [1]. Nowadays, efficient copper-catalyzed *enantioselective* allylic substitutions, which are promoted in the presence of appropriate chiral ligands, such as chiral phosphoramidite [2], amino acid-based chiral ligands [3], and so on [4], have been also established, and hence allylic substitution reactions have been recognized as one of the most important carbon–carbon bond forming reactions that deliver synthetically versatile organic molecules.

In analogy with the allylic substitutions in the *nonfluorinated* chemistry, the reactions of organocopper reagents toward *fluorinated* allylic halides or allylic alcohol derivatives have been published by several research groups thus far [5]. Very recently, our research group has also reported that the allylic substitution toward γ -fluoroallyl phosphates **1** using various cuprates derived from organolithium reagents, Grignard reagents, and organozinc reagents proceeded very smoothly to give the corresponding

γ -adducts **2** in a highly regioselective manner (Scheme 2) [6]. In addition, we have also found that the use of optically pure γ -fluoroallyl phosphate **3** derived from (*S*)-ethyl lactate, led to the highly regio- and diastereoselective reaction, the corresponding γ -product **4** with high optical purity being obtained in high yield.

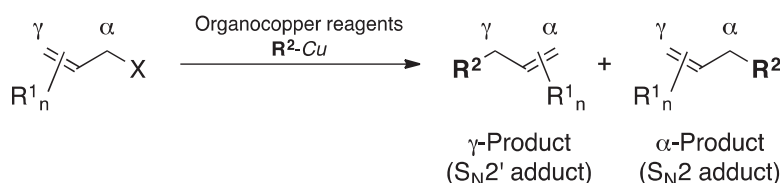
As a part of our research project on allylic substitution reactions of fluorinated substrates, herein is described the allylic substitutions of various organocopper reagents toward β -fluoroallyl phosphates **5** in detail (Scheme 3) [7]. In addition, we wish to disclose the synthesis of cyclic fluoroalkenes **7** via the ring-closing metathesis of thus obtained allylic substitution adducts **6** as well.

2. Results and discussion

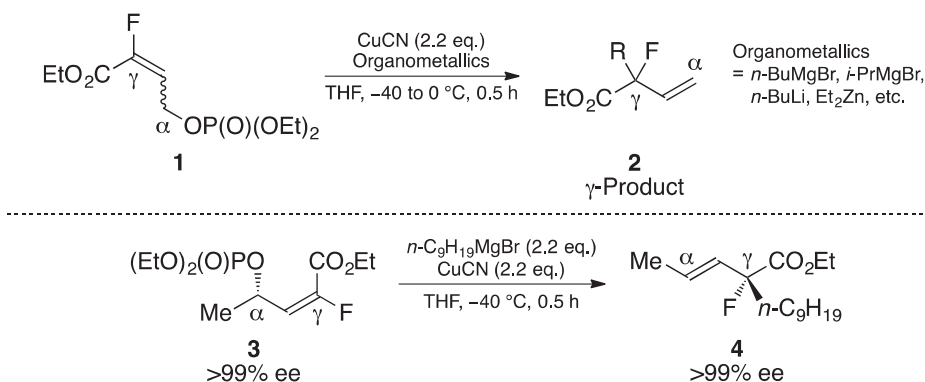
2.1. S_N2' reaction

Initially, we began with the preparation of various β -fluoroallyl phosphates, as shown in Scheme 4. Thus, treatment of various aldehydes with 1.2 equiv. of Horner–Wadsworth–Emmons reagent **8** in the presence of 1.2 equiv. of *n*-BuLi in THF at -78 °C gave the corresponding (*E*)-alkenes **9** in a highly stereoselective manner (*E*/*Z* = >95/5) [8,9]. Then, **9** were subjected to DIBAL-H-reduction [10] and the following phosphorylation [5], (*Z*)- β -fluoroallyl phosphates **10** being obtained in good yields. On the other hand, the reductive coupling of benzaldehyde (*R* = Ph) with $\text{Br}_2\text{FCCO}_2\text{Et}$

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Scheme 1. Allylic substitution reactions.

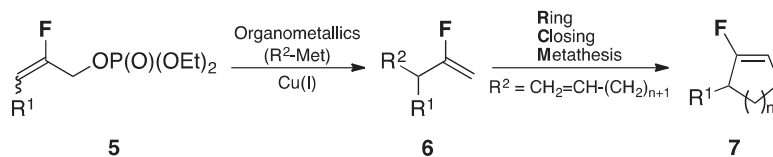
Scheme 2. Allylic substitution reactions of γ -fluoroallyl phosphates.

under the influence of CrCl_2 in THF at room temperature afforded the corresponding alkene **11a** in a highly *Z*-selective manner (*E/Z* = 5/ > 95) [11–13]. This alkene could be easily converted into (*Z*)- β -fluoroallyl phosphate **12a** in the same manner as utilized in the preparation of **10**.

With various types of substrates in hand, our attention was directed toward the examination of the allylic substitution reactions with **10**. First of all, we started the exploitation of the optimum reaction conditions by using **10a** ($\text{R} = \text{Ph}$), as shown in Table 1. Thus, treatment of **10a** with 1.1 equiv. of *n*-BuMgBr in the presence of 1.1 equiv. of CuCN in THF at -40°C for 1 h resulted in a complete recovery of the starting phosphate (Entry 1). Addition of TMSCl did not bring about any dramatic change in the yield (Entry 2). Very surprisingly, raising the amount of both *n*-BuMgBr and

CuCN from 1.1 to 2.2 equivalents led to a significant improvement of the yield, the desired $\text{S}_{\text{N}}2'$ adduct **13a** being obtained quantitatively (Entry 3). In this case, no other regioisomer **14a** was detected at all. It was also revealed that the reaction completed within 15 min (Entry 4). CuCN was found to be the copper salt of choice (Entries 5 and 6). Disappointingly, a Cu-catalyzed reaction did not proceed, the starting material being recovered quantitatively (Entry 7).

With the best reaction conditions in hand, we next examined the allylic substitution reactions of various organocopper reagents toward **10a** (Table 2, Entries 1–12). Various primary alkyl Grignard reagents, such as 3-buten-1-yl, 4-penten-1-yl, 5-hexen-1-yl, and 6-hepten-1-ylmagnesium bromides, could participate in the reaction very well to give the corresponding $\text{S}_{\text{N}}2'$ adducts in high



Scheme 3. Present work.

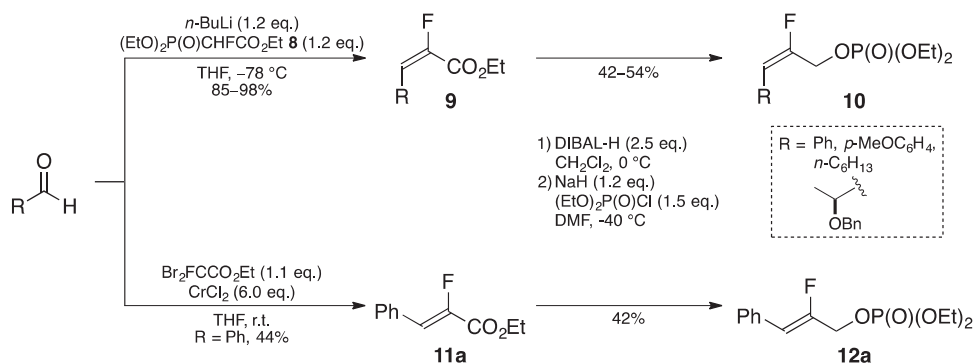
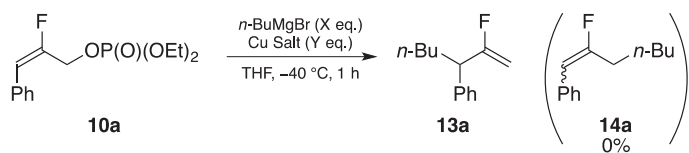
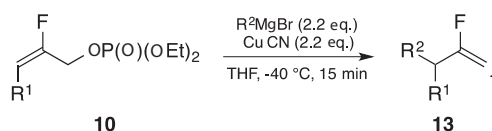
Scheme 4. Preparation of various β -fluoroallyl phosphates.

Table 1

Investigation of the reaction conditions

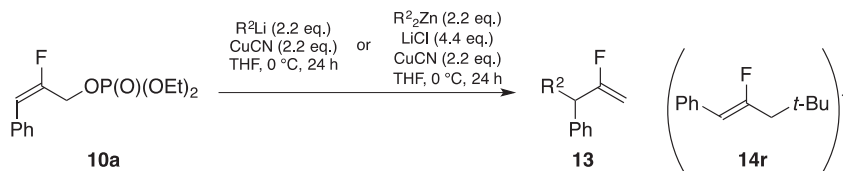


Entry	X/equiv.	Cu(I) Salt	Y/equiv.	Yield ^a / % of 13a	Recovery ^a / % of 10a
1	1.1	CuCN	1.1	0	Quant.
2 ^b	1.1	CuCN	1.1	0	Quant.
3	2.2	CuCN	2.2	Quant.	0
4 ^c	2.2	CuCN	2.2	Quant.	0
5 ^c	4.4	CuBr	2.2	88	0
6 ^c	4.4	CuI	2.2	72	0
7 ^d	2.0	CuCN	0.1	0	Quant.

^a Determined by ^{19}F NMR.^b TMSCl was used as an additive.^c Stirred for 15 min.^d Stirred for 16 h.**Table 2**Scope and limitation in the $\text{S}_{\text{N}}2'$ reaction

Entry	R ¹	R ²	Substrate	Yield ^a / % of 13	Product	Recovery ^a / % of 10
1	Ph	<i>n</i> -Bu	10a	Quant. (100)	13a	0
2	Ph		10a	80 (67)	13b	0
3	Ph		10a	Quant.	13c	0
4	Ph		10a	78 (50)	13d	0
5	Ph		10a	85 (77)	13e	0
6	Ph		10a	Quant. (95)	13f	0
7 ^b	Ph	Me	10a	51 (12) ^c	13g	39
8	Ph	<i>i</i> -Pr	10a	Quant. (54) ^c	13h	0
9	Ph	<i>c</i> -Hex	10a	Quant. (93)	13i	0
10 ^b	Ph	Bn	10a	0	13j	62
11 ^b	Ph		10a	0	13k	75
12 ^b	Ph		10a	0	13l	88
13	<i>p</i> -MeOC ₆ H ₄	<i>n</i> -Bu	10b	Quant. (96)	13m	0
14 ^d	Ph	<i>n</i> -Bu	12a	90	13a	0
15	<i>n</i> -C ₆ H ₁₃		10c	(89)	13n	0
16		<i>n</i> -Bu	10d	Quant. (89) ^e	13o	0
17 ^b	Ph	Ph	10a	9	13p	45
18 ^{b,f}	Ph	Ph	10a	41	13p	22

^a Determined by ^{19}F NMR. Values in parentheses are of isolated yield.^b Stirred for 24 h.^c The products were isolated in very low yields due to their volatility.^d The *Z* isomer was used.^e The diastereomeric ratio of the $\text{S}_{\text{N}}2'$ product was 95:5.^f In order to prepare the cuprate, 2.2 equiv. of CuI and 4.4 equiv. of PhMgBr were used.

Table 3 S_N2' reaction using organolithium or organozinc reagents

Entry	Organometallic	Yield ^a / % of 13	Product	Recovery ^a / % of 10a
1	MeLi	0	13g	94
2	<i>n</i> -BuLi	86 (79)	13a	10
3 ^b	<i>s</i> -BuLi	Quant. (90) ^c	13q	0
4 ^b	<i>t</i> -BuLi	18 ^d	13r	0
5	PhLi	0	13p	94
6	<i>p</i> -MeOC ₆ H ₄ Li	0	13s	94
7	<i>p</i> -CF ₃ C ₆ H ₄ Li	0	13t	68
8	Me ₂ Zn	0	13g	82
9 ^e	Et ₂ Zn	74 (32) ^f	13u	0
10	(<i>i</i> -Pr) ₂ Zn	93 ^f	13h	0
11	Ph ₂ Zn	0	13p	82

^a Determined by ¹⁹F NMR. Values in parentheses are of isolated yield.^b Stirred for 15 min at −20 °C.^c The diastereomeric ratio of the product was 64:36.^d The regioisomer **14r** was obtained in 43% yield as a major product.^e Stirred for 12 h.^f The products were isolated in very low yields due to their volatility.

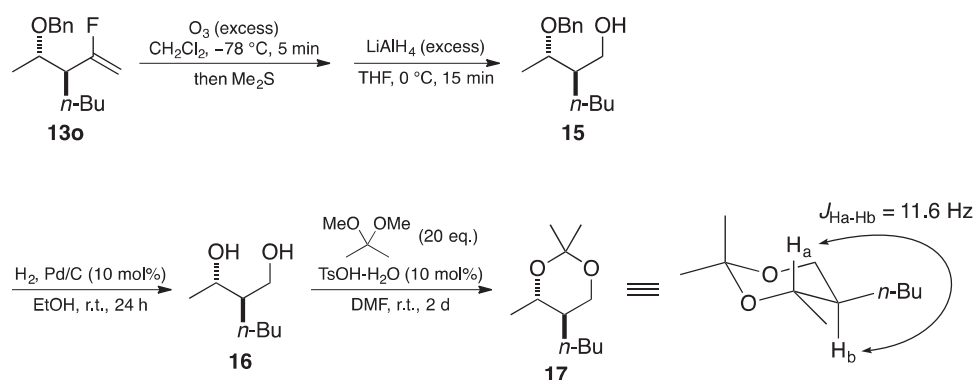
yields as a single isomer (Entries 2–5). Grignard reagent, derived from the alkyl halide having a dioxolanyl group at the terminus, was also found to be applicable (Entry 6), whereas the reaction with MeMgBr proceeded reluctantly to afford the γ -adduct in only 51% yield (Entry 7) [14]. Secondary alkyl Grignard reagents, such as *i*-PrMgBr and cyclohexylMgBr, are found to be good for the allylic substitution reactions (Entries 8 and 9), but less nucleophilic BnMgBr and AllylMgBr, in addition to vinylMgBr were all inert, the starting material being recovered in 62–88% yield (Entries 10–12) [14].

Then, we also examined the allylic substitution reactions of *n*-BuMgBr toward various allylic phosphates **10**. As shown in Entries 13 and 15, any influence of the substituents R¹ was not observed at all, the desired γ -adducts being provided in high yields. In addition, the geometry at the olefinic bond of **10** did not affect the reaction appreciably (Entry 14). We were very pleased that the reaction with **10d** having an asymmetric carbon center at δ position took place in a highly *anti*-selective manner (Entry 16). Arylcuprates were found to be much less reactive (Entry 17) [14], and the expected γ -adduct was given in only 41% even when higher-ordered cuprate, Ph₂Cu(MgBr)·(MgBrI) was used as an organo-copper reagent (Entry 18).

We also investigated the reaction with various organocopper reagents derived from organolithium or organozinc reagents instead of Grignard reagent as shown in Table 3. Generally, organocopper species derived from organolithium or organozinc reagents were less reactive than those derived from Grignard reagents, and hence all reactions were carried out at 0 °C. In analogy with the reaction using Grignard reagents, neither Me₂Cu(CN)Li₂ nor Aryl₂Cu(CN)Li₂ gave any expected products (Entries 1, 5, 6, and 7) [14]. The use of primary and secondary alkyllithium led to γ -products in high yields (Entries 2 and 3) whereas α -product **14r** was obtained as a major isomer in the case of *t*-BuLi (Entry 4). The same trend in the reactivity was observed in organozinc reagents. Thus, dialkylzinc reagents, except for Me₂Zn, reacted smoothly with **10a** in the presence of CuCN (Entries 8, 9, and 10). Again, Ph₂Zn could not participate in the reaction at all (Entry 11) [14].

2.2. Reaction mechanism for high anti-stereoselection in the S_N2' reaction with **10d**

The stereochemical assignment of **13o** was made as follows (Scheme 5). The ozonolysis of **13o** and the subsequent reduction

**Scheme 5.** Determination of the stereochemistry.

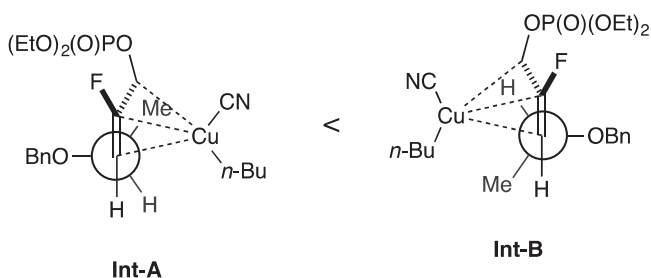


Fig. 1. Reaction mechanism for high *anti* stereoselection.

with LiAlH_4 gave the corresponding alcohol **15**. Hydrogenation of **15** afforded the corresponding 1,3-diol **16** which was treated with dimethoxypropane and a catalytic amount of TsOH to give the acetonide **17** [12,13]. Careful analysis of the ^1H NMR spectrum of **17** established the vicinal coupling constant between H_a and H_b to be 11.6 Hz, strongly suggesting its *anti* relative stereochemistry. In consequence, the stereochemistry of **13o** was unambiguously determined as *anti*.

The observed high *anti* diastereoselectivity can be explained by Felkin–Ahn model (Fig. 1) [15]. Thus, intermediates **Int-A** and **Int-B**, in which a benzyloxy group occupies a perpendicular position to the double bond is more stable than the other conformers, because the low-lying $\sigma_{\text{C-O}}^*$ orbital is aligned parallel with the π electron density by hyperconjugation from the reaction center toward the benzyloxy group. Then, the copper reagent preferably attacks the double bond, avoiding both the benzyloxy and the methyl groups due to a large steric hindrance. Accordingly, the reaction proceeds via **Int-B** preferentially, giving the corresponding *anti*-adduct **13o** as the major isomer.

2.3. Synthesis of monofluorinated carbocycles

As synthetic applications, we attempted the ring-closing metathesis [16] using thus obtained allylic substitution products. The results are summarized in Table 4 and Table 5. Generally, the ring-closing metathesis of fluorinated diene requires a high temperature according to the reported procedures [17]. Therefore, all reactions were carried out in a sealed tube. Thus, treatment of the terminal diene **13c** with 5 mol% of Grubbs 2nd generation

catalyst in CH_2Cl_2 at 0.01 concentration at 100°C for 2 h gave the expected cyclic adduct **18c** in 63% yield, together with 37% of the dimer **19c** (Table 4, Entry 1). Raising the catalyst quantity from 5 to 10 mol% led to a slight increase of the yield (Entry 2). Higher concentration caused a significant increase of the dimer **19c** (Entry 3). When the catalyst quantity was increased up to 20 mol%, the desired adduct was obtained in the best yield, without any trace of the dimer (Entry 4). The reaction in other solvents, like benzene or toluene, did not bring about satisfactory results (Entries 5 and 6).

With the optimum reaction conditions (Entry 4 in Table 4), we next carried out the ring-closing metathesis using various terminal dienes (Table 5).

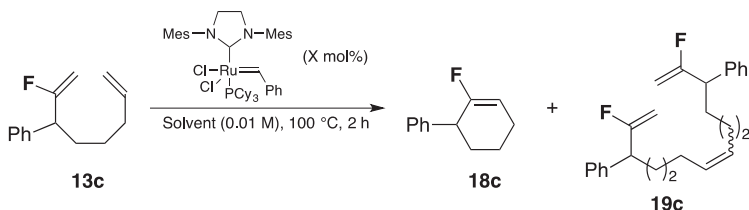
In analogy with **13c**, **13n** having an alkyl side chain underwent the smooth ring-closing metathesis, the corresponding **18n** being obtained quantitatively (Entry 2). Interestingly, a decrease of the catalyst quantity from 20 to 10 mol% did not bring about any significant change in the yield (Entry 3), whereas the reaction at higher concentration led to a significant increase of the dimer (Entry 4). It was noteworthy that only 1,7-dienes gave the corresponding cyclic adducts in high yields, and that the 7-membered ring compound **18d** was afforded in only 19% yield (Entry 6). Additionally, 1,6-diene **13b** and 1,9-diene **13e** gave the corresponding dimers almost quantitatively (Entries 5 and 7) [18].

We also examined the ring-closing metathesis of **21**, which was prepared via hydrolysis of the allylic substitution product **13f** and the following Grignard reaction using vinyl Grignard reagent toward the resulting aldehyde **20** (Scheme 6). Very disappointingly, the ring-closing metathesis of **21** caused an isomerization of the double bond, the corresponding ketone **22** being obtained in high yield. Therefore, we attempted the ring-closing reaction of the hydroxyl group-protected **23**. As a result, the ring-closing metathesis took place very smoothly to give the corresponding adduct **24** in high yield as a diastereomeric mixture.

3. Conclusion

In summary, we have developed the novel allylic substitution reactions of β -fluoroallyl phosphates with organocopper reagents. It was revealed that various types of β -fluoroallyl phosphates reacted smoothly with organoalkylcuprates, *not* organoarylcuprates, γ -products being obtained in a highly regioselective manner. Cuprates prepared from Grignard reagents and Cu(I) salt were

Table 4
Investigation of the ring-closing metathesis



Entry ^a	X/mol%	Solvent	Yield ^b /% of 18c	Yield ^b /% of 19c	Recovery ^b /% of 13c
1	5	CH_2Cl_2	63	37	0
2	10	CH_2Cl_2	71	29	0
3 ^c	10	CH_2Cl_2	43	57	0
4	20	CH_2Cl_2	80 (38) ^d	0	0
5	20	Benzene	43	36	9
6	20	Toluene	23	32	34

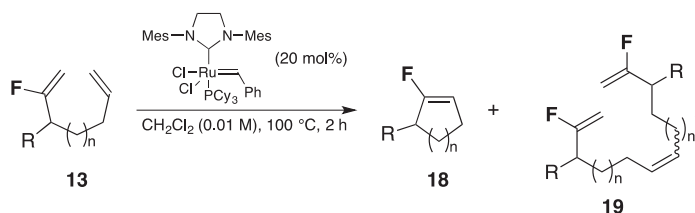
^a All reactions were carried out in a sealed tube.

^b Determined by ^{19}F NMR and GC. Value in parentheses is of isolated yield.

^c Carried out in a 0.05 M concentration.

^d The product was difficult to be separated from a small amount of other byproducts.

Table 5
Ring-closing metathesis of various fluorinated dienes



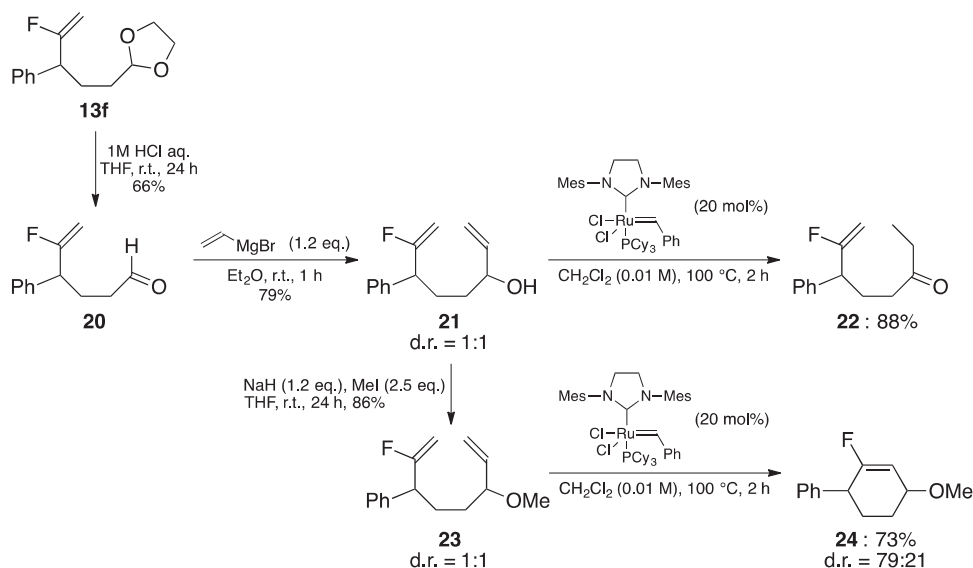
Entry ^a	R	n	Substrate	Yield ^b / % of 18	Product	Yield ^b / % of 19
1	Ph	2	13c	80 (38)	18c	0
2	<i>n</i> -C ₆ H ₁₃	2	13n	Quant. (100)	18n	0
3 ^c	<i>n</i> -C ₆ H ₁₃	2	13n	Quant.	18n	0
4 ^d	<i>n</i> -C ₆ H ₁₃	2	13n	54	18n	38
5	Ph	1	13b	0	18b	86
6	Ph	3	13d	19	18d	33
7	Ph	4	13e	Trace	18e	94

^a All reactions were carried out in a sealed tube.

^b Determined by ¹⁹F NMR and GC. Values in parentheses are of isolated yield.

^c 10 mol% of the catalyst was used.

^d Carried out in a 0.05 M concentration.



Scheme 6. Ring-closing metathesis of **21** and **23**.

found to be much more reactive than those derived from organolithium or organozinc reagents. As synthetic applications, we attempted the ring-closing metathesis of allylic substitution products. As a result, only 1,7-dienes could cyclize smoothly to give the corresponding fluorinated cyclohexene derivatives in high yields. On the other hand, 1,6-, 1,8-, and 1,9-dienes underwent an intermolecular metathesis to afford the corresponding dimers.

4. Experimental

4.1. General

Infrared spectra (IR) were recorded on a Shimadzu FTIR-8200A (PC) spectrophotometer and Thermo Electron Corp AVATAR 370DTGS spectrophotometer. ¹H NMR (500.13 MHz) and ¹³C NMR (125.75 MHz) spectra were measured with a Bruker DRX500 spectrometer or a JEOL JNM-AL400 spectrometer in a chloroform-*d* (CDCl_3) solution with tetramethylsilane (TMS) as an

internal reference. ¹⁹F NMR spectra were measured with a Bruker DPX300 (282.38 MHz) spectrometer or a JEOL JNM-AL400 spectrometer (376.05 MHz) in a CDCl_3 solution containing CFCl_3 as an internal reference. A JEOL JNM-EX90A (84.21 MHz, FT) spectrometer was used for determining the yields of the products with internal hexafluorobenzene (C_6F_6), trifluoromethylbenzene (BTF), or ethyl trifluoroacetate (TFA). It was also used for the determining regioselectivity and stereoselectivity. High-resolution mass spectra (HRMS) were measured with a JEOL JMS-700 mass spectrometer by electron impact (EI), chemical ionization (CI), or fast atom bombardment (FAB) method.

4.1.1. Materials

All chemicals were reagent-grade, and if necessary, were purified in the usual manner prior to use. LiBr-free MeLi, that is to say, "Methyl lithium, in diethyl ether (Br-free)" (abt 1 mol/L) was purchased from Kanto Chemical Co., Inc. Thin layer chromatography (TLC) was done with Merck 25 aluminum sheets (silica gel 60

F₂₅₄). Column chromatography was carried out with Wakogel C-200. All reactions were carried out under argon atmosphere.

4.2. Preparation of α -fluoro- α,β -unsaturated esters

Method A: To a solution of triethyl 2-fluoro-2-phosphonoacetate (1.45 g, 6 mmol) in THF (25 mL) was added *n*-BuLi (1.6 M in *n*-hexane, 3.78 mL, 6 mmol) at -78°C . After stirring of the reaction mixture for 30 min at that temperature, benzaldehyde (0.53 g, 5 mmol) was added to the reaction mixture, and the resulting mixture was stirred at -78°C for 30 min. The reaction was quenched with diluted HCl aq., and the whole was extracted with diethyl ether. The combined organic layers were dried over anhydrous Na₂SO₄, filtered, and concentrated *in vacuo*. The residue was purified by silica gel column chromatography to give (*E*)-ethyl 2-fluoro-3-phenyl-2-propenoate (**9a**) (0.95 g, 98% yield).

Method B: To a suspension of CrCl₂ powder (95%, 0.78 g, 6.0 mmol) in anhydrous THF (8 mL) was added a mixture of ethyl dibromofluoroacetate (0.29 g, 1.1 mmol) and benzaldehyde (0.106 g, 1.0 mmol) in anhydrous THF (2 mL) at room temperature under an argon atmosphere. The resulting mixture was stirred at that temperature for 30 min, and then the reaction was quenched with water. The reaction mixture was filtered, and the filtrate was extracted with diethyl ether three times. The combined organic layers were dried over anhydrous Na₂SO₄, filtered, and concentrated *in vacuo*. The residue was purified by silica gel column chromatography to give (*Z*)-ethyl 2-fluoro-3-phenyl-2-propenoate (**11a**) (0.085 g, 44% yield).

4.2.1. (*E*)-Ethyl 2-fluoro-3-phenyl-2-propenoate (**9a**)

Known compound; ¹H NMR (CDCl₃) 1.25 (3H, t, *J* = 7.2 Hz), 4.26 (2H, q, *J* = 7.2 Hz), 6.92 (1H, d, *J* = 22.2 Hz), 7.32–7.38 (3H, m), 7.46–7.48 (2H, m) ppm; ¹³C NMR (CDCl₃) 13.71, 61.47, 121.28 (d, *J* = 25.9 Hz), 127.92, 128.57, 129.55 (d, *J* = 2.8 Hz), 130.92 (d, *J* = 9.3 Hz), 146.91 (d, *J* = 255.4 Hz), 160.36 (d, *J* = 36.1 Hz) ppm; ¹⁹F NMR (CDCl₃) δ –117.76 (1F, d, *J* = 22.2 Hz) ppm; IR (neat) 2984, 1732, 1656, 1447, 1396, 1347, 1229, 1229, 1131, 1023, 777, 752, 694, 643 cm^{–1}; HRMS (FAB) Calcd for C₁₁H₁₁FO₂ (M⁺) 194.0743, found 194.0753.

4.2.2. (*E*)-Ethyl 2-fluoro-3-(4-methoxyphenyl)-2-propenoate (**9b**)

Known compound; ¹H NMR (CDCl₃) 1.30 (3H, t, *J* = 6.8 Hz), 3.82 (3H, m), 4.28 (2H, q, *J* = 6.8 Hz), 6.85 (1H, d, *J* = 24.0 Hz), 6.88 (2H, d, *J* = 8.8 Hz), 7.51 (2H, d, *J* = 8.8 Hz) ppm; ¹³C NMR (CDCl₃) δ 13.99, 55.27, 61.49, 113.51, 121.96 (d, *J* = 26.4 Hz), 123.01 (d, *J* = 9.9 Hz), 132.65 (d, *J* = 3.3 Hz), 145.78 (d, *J* = 252.1 Hz), 160.11, 160.76 (d, *J* = 35.5 Hz) ppm; ¹⁹F NMR (CDCl₃) –119.77 (d, *J* = 24.4 Hz) ppm; IR (neat) 2982, 2983, 2908, 2840, 1728, 1606, 1512, 1465, 1396, 1377, 1350, 1303, 1227, 1179, 1130, 1100, 1031, 902, 827, 773 cm^{–1}; HRMS (FAB) Calcd for C₁₂H₁₃FO₃ (M⁺) 224.0849, found 224.0857.

4.2.3. (*E*)-Ethyl 2-fluoro-2-nonen-1-enoate (**9c**)

¹H NMR (CDCl₃) 0.89 (3H, t, *J* = 6.4 Hz), 1.25–1.50 (8H, m), 1.35 (3H, t, *J* = 7.2 Hz), 2.47–2.55 (2H, m), 4.30 (2H, q, *J* = 7.2 Hz), 5.92 (1H, dt, *J* = 21.1, 8.0 Hz) ppm; ¹³C NMR (CDCl₃) 14.03, 14.10, 22.54, 25.44 (d, *J* = 4.9 Hz), 28.82, 29.16 (d, *J* = 2.4 Hz), 31.53, 61.24, 123.85 (d, *J* = 18.2 Hz), 146.91 (d, *J* = 249.6 Hz), 161.07 (d, *J* = 36.3 Hz) ppm; ¹⁹F NMR (CDCl₃) –123.32 (d, *J* = 21.1 Hz) ppm; IR (neat) 2929, 2859, 1730, 1665, 1466, 1376, 1348, 1326, 1236, 1146, 1120, 1024, 896, 858, 777 cm^{–1}; HRMS (FAB) Calcd for C₁₁H₁₉FO₂ (M⁺) 202.1369, found 202.1375.

4.2.4. (*2E*, *3S*)-Ethyl 2-fluoro-4-benzyloxy-2-pentenoate (**9d**)

¹H NMR (CDCl₃) δ 1.32 (3H, t, *J* = 7.1 Hz), 1.35 (3H, dd, *J* = 1.0, 6.3 Hz), 4.28 (1H, q, *J* = 7.1 Hz), 4.29 (1H, q, *J* = 7.1 Hz), 4.43 (1H, d, *J* = 1.7 Hz), 4.54 (1H, d, *J* = 11.7 Hz), 4.95–5.10 (1H, m), 5.91 (1H, dd,

J = 8.9, 20.7 Hz), 7.25–7.34 (5H, m); ¹³C NMR (CDCl₃) δ 14.00, 21.22 (d, *J* = 2.8 Hz), 61.74, 69.50 (d, *J* = 7.4 Hz), 70.93, 125.53 (d, *J* = 14.8 Hz), 127.65, 127.69, 128.38, 138.16, 147.28 (d, *J* = 259.5 Hz), 160.54 (d, *J* = 36.0 Hz); ¹⁹F NMR (CDCl₃) –120.55 (1F, d, *J* = 20.7 Hz) ppm; IR (neat) 3449, 2370, 2345, 2250, 2185, 2111, 1121, 617 cm^{–1}; HRMS (FAB) Calcd for C₁₄H₁₈FO₃ (M + H) 253.1240, found 253.1240.

4.2.5. (*Z*)-Ethyl 2-fluoro-3-phenylpropenoate (**11a**)

Known compound; ¹H NMR (CDCl₃) 1.38 (3H, t, *J* = 7.1 Hz), 4.35 (2H, q, *J* = 7.1 Hz), 6.92 (1H, d, *J* = 35.2 Hz), 7.37–7.40 (3H, m), 7.64–7.65 (2H, m) ppm; ¹³C NMR (CDCl₃) 14.19, 61.85, 117.45 (d, *J* = 4.91 Hz), 128.78, 129.65 (d, *J* = 2.6 Hz), 130.26 (d, *J* = 8.2 Hz), 131.13 (d, *J* = 4.3 Hz), 147.00 (d, *J* = 267.6 Hz), 161.41 (d, *J* = 35.2 Hz) ppm; ¹⁹F NMR (CDCl₃) –125.81 (1F, d, *J* = 36.9 Hz) ppm.

4.3. Preparation for allyl alcohols

To a CH₂Cl₂ (12 mL) solution of (*E*)-ethyl 2-fluoro-3-phenylpropenoate (0.95 g, 5 mmol) was added diisobutylaluminum hydride (1 M in *n*-hexane, 13 mL, 12.5 mmol) at 0 °C. After stirring for 30 min, the reaction was quenched with diluted HCl aq. and the whole was extracted with CH₂Cl₂. The combined organic layers were dried over anhydrous Na₂SO₄, concentrated *in vacuo*. The residue was purified by silica gel column chromatography to give (*E*)-2-fluoro-3-phenyl-2-propen-1-ol (0.72 g, 95% yield).

4.3.1. (*E*)-2-Fluoro-3-phenyl-2-propen-1-ol

Known compound; ¹H NMR (CDCl₃) 2.66 (1H, s), 4.35 (2H, d, *J* = 22.2 Hz), 6.39 (1H, d, *J* = 20.0 Hz), 7.23–7.29 (3H, m), 7.33–7.36 (2H, m) ppm; ¹³C NMR (CDCl₃) 58.08 (d, *J* = 29.2 Hz), 111.38 (d, *J* = 25.9 Hz), 127.37, 128.52, 128.55 (d, *J* = 2.6 Hz), 132.98 (d, *J* = 12.5 Hz), 159.53 (d, *J* = 253.9 Hz) ppm; ¹⁹F NMR (CDCl₃) δ –109.68 (1F, dt, *J* = 20.0, 22.2 Hz) ppm; HRMS Calcd for C₉H₉FO (M⁺) 152.0637, found 152.0642.

4.3.2. (*E*)-2-Fluoro-3-(4-methoxyphenyl)-2-propen-1-ol

Known compound; ¹H NMR (CDCl₃) 2.13 (1H, s), 3.18 (3H, s), 4.35 (2H, d, *J* = 22.4 Hz), 6.34 (1H, d, *J* = 20.0 Hz), 6.88 (2H, d, *J* = 8.8 Hz), 7.12 (2H, d, *J* = 8.8 Hz) ppm; ¹³C NMR (CDCl₃) 55.25, 58.34 (d, *J* = 29.7 Hz), 111.04 (d, *J* = 26.5 Hz), 114.03, 125.24 (d, *J* = 12.3 Hz), 129.75, 158.68 (d, *J* = 252.1 Hz), 158.96 ppm; ¹⁹F NMR (CDCl₃) –111.11 (1F, dt, *J* = 20.0, 22.4 Hz) ppm; HRMS (FAB) Calcd for C₁₀H₁₁FO₂ (M⁺) 182.0743, found 182.0742.

4.3.3. (*E*)-2-Fluoro-2-nonen-1-ol

Known compound; ¹H NMR (CDCl₃) 0.88 (3H, t, *J* = 6.0 Hz), 1.27–1.38 (8H, m), 1.71 (1H, s), 2.00 (2H, dt, *J* = 7.9, 7.9 Hz), 4.23 (2H, d, *J* = 20.8 Hz), 5.21 (1H, dt, *J* = 21.6, 7.9 Hz) ppm; ¹³C NMR (CDCl₃) δ 14.03, 22.55, 25.18 (d, *J* = 8.2 Hz), 28.63, 29.84 (d, *J* = 2.4 Hz), 31.56, 57.16 (d, *J* = 30.6 Hz), 109.33 (d, *J* = 19.0 Hz), 157.47 (d, *J* = 245.5 Hz) ppm; ¹⁹F NMR (CDCl₃) –115.49 (1F, dt, *J* = 21.6, 20.8 Hz) ppm; IR (neat) 3337, 2956, 2928, 2858, 1700, 1466, 1379, 1241, 1159, 1110, 1028, 862, 734 cm^{–1}; HRMS (EI) Calcd for C₉H₁₇FO (M⁺) 160.1263, found 160.1256.

4.3.4. (*2E*, *3S*)-3-Benzyloxy-2-fluoro-2-penten-1-ol

¹H NMR (CDCl₃) 1.33 (3H, dd, *J* = 6.3, 1.0 Hz), 1.89 (1H, s), 4.11–4.24 (3H, m), 4.39 (1H, d, *J* = 12.0 Hz), 4.57 (1H, dd, *J* = 12.0 Hz), 5.23 (dd, 1H, *J* = 9.4, 20.5 Hz), 7.26–7.37 (5H, m) ppm; ¹³C NMR (CDCl₃) 22.22, 57.68 (d, *J* = 30.5 Hz), 69.21 (d, *J* = 12.0 Hz), 69.79, 111.30 (d, *J* = 17.6 Hz), 127.70, 128.49, 138.08, 159.64 (d, *J* = 254.9 Hz) ppm; IR (neat) 3403, 3032, 2976, 2929, 2929, 2871, 1699, 1454, 1375, 1292, 1169, 1100, 1072, 1028, 868, 833, 740, 698 cm^{–1}.

4.3.5. (Z)-2-Fluoro-3-phenyl-2-propen-1-ol

^1H NMR (CDCl_3) 2.19 (1H, s), 4.26 (2H, d, $J = 14.4$ Hz), 5.77 (1H, d, $J = 38.8$ Hz), 7.23–7.51 (5H, m) ppm; ^{13}C NMR (CDCl_3) 61.83 (d, $J = 32.4$ Hz), 107.46 (d, $J = 6.5$ Hz), 127.47 (d, $J = 1.6$ Hz), 128.47, 128.66 (d, $J = 7.2$ Hz), 132.62 (d, $J = 2.9$ Hz), 158.07 (d, $J = 266.8$ Hz) ppm; ^{19}F NMR (CDCl_3) –113.8391 to –113.72 (1F, m) ppm; IR (neat) 3357, 3057, 2972, 2927, 2387, 1691, 1493, 1449, 1348, 1225, 1160, 1019, 877, 751, 693, 538 cm^{-1} ; HRMS (EI) Calcd for $\text{C}_9\text{H}_9\text{FO}$ (M^+) 152.0637, found 152.0638.

4.4. Preparation of allylphosphates

To a suspension of NaH (60% in oil, 0.24 g, 6 mmol) in DMF (12.5 mL) was added (*E*)-2-fluoro-3-phenyl-2-propen-1-ol (0.72 g, 5 mmol) at 0 °C. After stirring for 20 min, the reaction mixture was cooled to –40 °C, and diethyl chlorophosphate (1.08 mL, 7.5 mmol) was added dropwise. The resulting mixture was stirred at –40 °C for 3 h. The reaction was quenched with diluted HCl aq. The whole was extracted with diethyl ether, and the combined organic layers were dried over anhydrous Na_2SO_4 , concentrated *in vacuo*. The residue was purified by silica gel column chromatography to give (*E*)-diethyl 2-fluoro-3-phenyl-2-propenylphosphate (0.92 g, 64% yield).

4.4.1. (*E*)-Diethyl 2-fluoro-3-phenyl-2-propen-1-yl phosphate (**10a**)

^1H NMR (CDCl_3) 1.33 (6H, t, $J = 7.1$ Hz), 4.13 (2H, q, $J = 7.1$ Hz), 4.15 (2H, q, $J = 7.1$ Hz), 4.73 (2H, dd, $J = 7.4$, 21.7 Hz), 6.52 (1H, d, $J = 18.9$ Hz), 7.25–7.37 (5H, m) ppm; ^{13}C NMR (CDCl_3) 15.86 (d, $J = 6.7$ Hz), 61.89 (d, $J = 4.8$ Hz), 63.03 (dd, $J = 6.3$, 23.3 Hz), 114.14 (d, $J = 25.0$ Hz), 127.73, 128.42, 128.55, 132.09 (d, $J = 11.7$ Hz), 155.27 (dd, $J = 7.5$, 247.2 Hz) ppm; ^{19}F NMR (CDCl_3) δ –108.38 (1F, dt, $J = 21.7$, 18.9 Hz) ppm; IR (neat) 3469, 3030, 2987, 2934, 2911, 1688, 1494, 1480, 1447, 1394, 1370, 1267, 1217, 1149, 1028, 890, 818, 752, 701, 630 cm^{-1} ; HRMS (FAB) Calcd for $\text{C}_{13}\text{H}_{18}\text{FO}_4\text{P}$ (M^+) 288.0927, found 288.0919.

4.4.2. (*E*)-Diethyl 2-fluoro-3-(4-methoxyphenyl)-2-propen-1-yl phosphate (**10b**)

^1H NMR (CDCl_3) 1.30–1.38 (6H, m), 3.80 (3H, s), 4.10–4.18 (4H, m), 4.72 (2H, dd, $J = 7.2$, 22.0 Hz), 6.47 (1H, d, $J = 18.8$ Hz), 6.88 (2H, d, $J = 8.8$ Hz), 7.20 (2H, d, $J = 8.8$ Hz) ppm; ^{13}C NMR (CDCl_3) 16.02 (d, $J = 6.6$ Hz), 55.25, 62.32 (dd, $J = 30.6$, 4.9 Hz), 64.05 (d, $J = 5.8$ Hz), 114.01 (d, $J = 28.1$ Hz), 114.15, 124.51 (d, $J = 11.6$ Hz), 129.78 (d, $J = 3.3$ Hz), 154.55 (dd, $J = 244.6$, 7.5 Hz), 159.29 ppm; ^{19}F NMR (CDCl_3) –109.93 (dt, $J = 18.8$, 22.0 Hz) ppm; IR (neat) 3369, 2985, 2937, 2910, 2840, 1683, 1609, 1513, 1463, 1394, 1254, 1150, 1032, 983, 893, 841, 821, 576 cm^{-1} ; HRMS (FAB) Calcd for $\text{C}_{14}\text{H}_{20}\text{FO}_5\text{P}$ (M^+) 318.1032, found 318.1034.

4.4.3. (*E*)-Diethyl 2-fluoro-2-nonen-1-yl phosphate (**10c**)

^1H NMR (CDCl_3) 0.86 (3H, t, $J = 6.4$ Hz), 1.25–1.34 (8H, m), 1.321 (3H, t, $J = 7.1$ Hz), 1.324 (3H, t, $J = 7.1$ Hz), 2.00–2.10 (2H, m), 4.05–4.15 (4H, m), 4.60 (2H, dd, $J = 8.4$, 21.6 Hz), 5.33 (1H, dt, $J = 20.0$, 8.0 Hz) ppm; ^{13}C NMR (CDCl_3) 13.98, 15.99 (d, $J = 7.4$ Hz), 22.51, 25.29 (d, $J = 7.5$ Hz), 28.58, 29.59 (d, $J = 2.4$ Hz), 31.52, 60.86 (dd, $J = 31.4$, 4.9 Hz), 63.99 (d, $J = 5.7$ Hz), 112.68 (d, $J = 19.0$ Hz), 153.75 (dd, $J = 6.5$, 247.2 Hz) ppm; ^{19}F NMR (CDCl_3) –113.54 (1F, dt, $J = 20.0$, 21.6 Hz) ppm; IR (neat) 2930, 2858, 1701, 1459, 1393, 1276, 1166, 1120, 1032, 874, 839, 752 cm^{-1} .

4.4.4. (2*E*, 3*S*)-Diethyl 2-fluoro-3-benzyloxy-2-penten-1-yl phosphate (**10d**)

^1H NMR (CDCl_3) 1.30–1.37 (9H, m), 4.07–4.14 (4H, m), 4.17–4.21 (1H, m), 4.37 (1H, $J = 11.8$ Hz), 4.55 (1H, d, $J = 11.8$ Hz), 4.56–4.65 (2H, m), 5.36 (1H, dd, $J = 9.8$, 19.4 Hz), 7.27–7.34 (5H, m) ppm; ^{13}C NMR (CDCl_3) 16.03 (d, $J = 5.9$ Hz), 22.15 (d, $J = 2.0$ Hz), 61.01

(dd, $J = 30.3$, 5.2 Hz), 64.08 (d, $J = 6.0$ Hz), 69.55 (d, $J = 10.9$ Hz), 70.08, 114.67 (d, $J = 16.2$ Hz), 127.65, 127.68, 128.45, 138.03, 155.50 (dd, $J = 6.8$, 256.2 Hz) ppm; ^{19}F NMR (CDCl_3) –109.06 (dt, $J = 22.2$, 19.6 Hz) ppm; IR (neat) 3064, 3031, 2933, 2910, 1702, 1454, 1392, 1273, 1175, 1030, 876, 849, 741, 699 cm^{-1} ; HRMS (FAB) Calcd for $\text{C}_{16}\text{H}_{25}\text{FO}_5\text{P}$ ($\text{M} + \text{H}$) 347.1424, found 347.1417.

4.4.5. (*Z*)-Diethyl 2-fluoro-3-phenylpropenylphosphate (**12a**)

^1H NMR (CDCl_3) 1.346 (3H, t, $J = 7.1$ Hz), 1.348 (3H, t, $J = 7.1$ Hz), 4.10–4.20 (4H, m), 4.67 (2H, dd, $J = 8.6$, 16.9 Hz), 5.88 (1H, d, $J = 37.1$ Hz), 7.27–7.53 (5H, m) ppm; ^{13}C NMR (CDCl_3) 16.03 (d, $J = 6.9$ Hz), 64.06 (d, $J = 6.1$ Hz), 65.71 (dd, $J = 5.3$, 32.0 Hz), 110.85 (d, $J = 6.5$ Hz), 128.01 (d, $J = 2.4$ Hz), 128.52, 128.90 (d, $J = 7.4$ Hz), 132.06 (d, $J = 3.0$ Hz), 153.88 (dd, $J = 7.2$, 266.5 Hz) ppm; ^{19}F NMR (CDCl_3) –133.48 (1F, dt, $J = 37.1$, 16.9 Hz) ppm; IR (neat) 3060, 2985, 2911, 1693, 1496, 1448, 1355, 1269, 1170, 1030, 886, 757, 695, 606 cm^{-1} ; HRMS (FAB) Calcd for $\text{C}_{13}\text{H}_{18}\text{FO}_4\text{P}$ (M^+) 288.0927, found 288.0923.

4.5. Allylic substitution reactions of β -fluoroallylic phosphate with organocuprates

Typical procedure for the preparation of Grignard reagent: To a suspension of magnesium (0.13 g, 5.32 mmol) in anhydrous THF (1 mL) was added dropwise 1-bromobutane (0.1 g, 0.71 mmol) at room temperature under an argon. After the reaction started, a THF solution of the remaining 1-bromobutane (0.57 g, 4.19 mmol/anhydrous THF 4 mL) was dropwise into the reaction mixture at that temperature. After stirring for 50 min, 0.8 M of *n*-BuMgBr in THF was obtained. (The concentration was determined by the titration experiment.)

Method A: To a suspension of CuCN (0.059 g, 0.66 mmol) in anhydrous THF (1 mL) was added dropwise a THF solution of *n*-BuMgBr (0.8 M, 0.83 mL, 0.66 mmol, prepared from *n*-BuBr and magnesium) at –78 °C under argon. After stirring for 5 min, the mixture was warmed to –10 °C, and then stirred for 10 min. The reaction mixture was again cooled to –40 °C, and to this mixture was added dropwise a solution of (*E*)-diethyl 2-fluoro-3-phenyl-2-propenylphosphate (0.086 g, 0.3 mmol) in anhydrous THF (2 mL). After 15 min, the reaction was quenched with an aqueous saturated NH_4Cl solution and the whole was extracted with diethyl ether. The combined organic layers were dried over anhydrous Na_2SO_4 , concentrated *in vacuo*. The residue was purified by silica gel column chromatography to give 2-fluoro-3-phenyl-1-heptene (**13a**) (quant., ^{19}F NMR yield).

Method B: To a suspension of CuI (0.126 g, 0.66 mmol) in anhydrous THF (1 mL) was added dropwise a THF solution of PhMgBr (1.05 M, 1.26 mL, 1.32 mmol, prepared from PhBr and magnesium) at –78 °C under argon. After stirring for 5 min, the mixture was warmed to –10 °C, then stirred for 15 min. The reaction mixture was again cooled to –78 °C, and to this mixture was added dropwise a solution of (*E*)-diethyl 2-fluoro-3-phenyl-2-propenylphosphate (0.086 g, 0.3 mmol) in anhydrous THF (2 mL). After stirring of the reaction mixture for 24 h, the reaction was quenched with an aqueous saturated NH_4Cl solution and the whole was extracted with diethyl ether. The combined organic layers were dried over anhydrous Na_2SO_4 , concentrated *in vacuo*. The residue was purified by silica gel column chromatography to give 2-fluoro-3,3-diphenyl-1-propene (**13r**) (41%, ^{19}F NMR yield).

Method C: To a suspension of CuCN (0.059 g, 0.66 mmol) in anhydrous THF (2 mL) was added dropwise a hexane solution of *n*-BuLi (1.6 M, 0.41 mL, 0.66 mmol) at –78 °C under an argon. After stirring for 10 min, the mixture was warmed to –20 °C, and then stirred for 15 min. The reaction mixture was again warmed to 0 °C, and to this mixture was added dropwise a solution of (*E*)-diethyl 2-fluoro-3-phenyl-2-propenylphosphate (0.086 g, 0.3 mmol) in

anhydrous THF (2 mL). After 24 h, the reaction was quenched with an aqueous saturated NH_4Cl solution and the whole was extracted with diethyl ether. The combined organic layers were dried over anhydrous Na_2SO_4 , concentrated *in vacuo*. The residue was purified by silica gel column chromatography to give 2-fluoro-3-phenyl-1-pentene (**13a**) (90%, ^{19}F NMR yield).

Method D: To a suspension of $\text{CuCN}\cdot 2\text{LiCl}$ (0.059 g, 0.66 mmol) in anhydrous THF (2 mL) was added dropwise a hexane solution of Et_2Zn (1.6 M, 0.41 mL, 0.66 mmol) at -78°C under an argon. After stirring for 10 min, the mixture was warmed to 0°C , and then stirred for 10 min. The reaction mixture was again cooled to 0°C , and to this mixture was added dropwise a solution of (*E*)-diethyl 2-fluoro-3-phenyl-2-propenylphosphate (0.086 g, 0.3 mmol) in anhydrous THF (2 mL). After 24 h, the reaction was quenched with an aqueous saturated NH_4Cl solution and the whole was extracted with diethyl ether. The combined organic layers were dried over anhydrous Na_2SO_4 , concentrated *in vacuo*. The residue was purified by silica gel column chromatography to give 2-fluoro-3-phenyl-1-heptene (**13w**) (74%, ^{19}F NMR yield).

4.5.1. 2-Fluoro-3-phenyl-1-heptene (**13a**)

Yield: 100%; ^1H NMR (CDCl_3) 0.87 (3H, t, $J = 6.3$ Hz), 1.20–1.35 (4H, m), 1.71–1.74 (1H, m), 1.89–1.92 (1H, m), 3.36 (1H, dt, $J = 17.8$, 7.7 Hz), 4.34 (1H, dd, $J = 2.8$, 50.1 Hz), 4.58 (1H, dd, $J = 2.8$, 17.8 Hz), 7.21–7.32 (5H, m) ppm; ^{13}C NMR (CDCl_3) 13.89, 22.50, 29.62, 32.30 (d, $J = 0.4$ Hz), 48.62 (d, $J = 25.2$ Hz), 90.06 (d, $J = 20.5$ Hz), 126.86, 127.84, 128.48, 141.01, 168.20 (d, $J = 260.5$ Hz) ppm; ^{19}F NMR (CDCl_3) –99.42 (1F, ddd, $J = 17.8$, 17.8, 50.1 Hz) ppm; IR (neat) 3030, 2932, 2862, 1667, 1496, 1454, 1258, 1182, 925, 849, 750, 698 cm^{-1} ; HRMS Calcd for $\text{C}_{12}\text{H}_{17}\text{F}$ (M^+) 192.1314, found 192.1309.

4.5.2. 2-Fluoro-3-phenyl-1,6-heptadiene (**13b**)

Yield: 67%; ^1H NMR (CDCl_3) 1.77–1.86 (1H, m), 1.97–2.06 (3H, m), 3.41 (1H, dt, $J = 17.6$, 7.1 Hz), 4.30 (1H, dd, $J = 2.8$, 50.4 Hz), 4.60 (1H, dd, $J = 2.8$, 17.6 Hz), 4.98 (1H, d, $J = 10.0$ Hz), 5.00 (1H, d, $J = 18.0$ Hz), 5.73–5.83 (1H, m), 7.16–7.33 (5H, m) ppm; ^{13}C NMR (CDCl_3) 31.30, 31.52 (d, $J = 3.3$ Hz), 47.69 (d, $J = 25.6$ Hz), 90.34 (d, $J = 20.7$ Hz), 115.30, 126.97, 127.87, 128.52, 137.74, 140.48, 167.75 (d, $J = 260.4$ Hz) ppm; ^{19}F NMR (CDCl_3) –99.77 (1F, ddd, $J = 17.6$, 17.6, 50.4 Hz) ppm; IR (neat) 3064, 3030, 3003, 2978, 2936, 2866, 1667, 1641, 1495, 1454, 1257, 1183, 916, 851 cm^{-1} ; HRMS (EI) Calcd for $\text{C}_{13}\text{H}_{15}\text{F}$ (M^+) 190.1158, found 190.1149.

4.5.3. 2-Fluoro-3-phenyl-1,7-octadiene (**13c**)

Yield: quant. (determined by ^{19}F NMR); ^1H NMR (CDCl_3) 1.30–1.48 (2H, m), 1.69–1.78 (1H, m), 1.88–1.97 (1H, m), 2.00–2.10 (2H, m), 3.38 (1H, dt, $J = 17.6$, 7.8 Hz), 4.35 (1H, dd, $J = 2.8$, 50.4 Hz), 4.60 (1H, dd, $J = 2.8$, 17.6 Hz), 4.94 (1H, dd, $J = 2.6$, 10.4 Hz), 5.01 (1H, dq, $J = 2.6$, 17.2 Hz), 5.76 (1H, ddt, $J = 17.2$, 10.4, 4.8 Hz), 7.22–7.34 (5H, m) ppm; ^{13}C NMR (CDCl_3) 26.67, 31.95 (d, $J = 3.4$ Hz), 33.49, 48.46 (d, $J = 25.4$ Hz), 90.15 (d, $J = 20.4$ Hz), 114.76, 126.92, 127.80, 128.50, 138.34, 140.75, 168.91 (d, $J = 260.2$ Hz) ppm; ^{19}F NMR (CDCl_3) –99.50 (1F, ddd, $J = 17.6$, 17.6, 50.4 Hz) ppm; IR (neat) 3064, 3030, 2936, 2864, 1667, 1454, 1183, 913, 850, 757, 699 cm^{-1} ; HRMS Calcd for $\text{C}_{14}\text{H}_{17}\text{F}$ (M^+) 204.1314, found 204.1321.

4.5.4. 2-Fluoro-3-phenyl-1,8-nonadiene (**13d**)

Yield: 50%; ^1H NMR (CDCl_3) 1.19–1.45 (4H, m), 1.67–1.76 (1H, m), 1.87–2.04 (3H, m), 3.36 (1H, dt, $J = 17.6$, 7.6 Hz), 4.34 (1H, dd, $J = 2.8$, 50.4 Hz), 4.59 (1H, dd, $J = 2.8$, 17.6 Hz), 4.90–5.00 (2H, m), 5.77 (1H, ddt, $J = 17.2$, 13.2, 7.2 Hz), 7.22–7.33 (5H, m) ppm; ^{13}C NMR (CDCl_3) 26.89, 28.68, 32.38, 33.52, 48.54 (d, $J = 25.6$ Hz), 90.16 (d, $J = 20.7$ Hz), 114.39, 126.89, 127.80, 128.49, 138.75, 140.86, 168.02 (d, $J = 241.7$ Hz) ppm; ^{19}F NMR (CDCl_3) –99.53 (1F, ddd,

$J = 17.6$, 17.6, 50.4 Hz) ppm; IR (neat) 3064, 3029, 2933, 2860, 1668, 1495, 1454, 1260, 1183, 994, 913, 850, 699 cm^{-1} ; HRMS (EI) Calcd for $\text{C}_{15}\text{H}_{19}\text{F}$ (M^+) 218.1471, found 218.1479.

4.5.5. 2-Fluoro-3-phenyl-1,9-decadiene (**13e**)

Yield: 77%; ^1H NMR (CDCl_3) 1.27–1.37 (6H, m), 1.66–1.75 (1H, m), 1.86–1.95 (1H, m), 1.98–2.03 (2H, m), 3.36 (1H, dt, $J = 17.6$, 5.6 Hz), 4.34 (1H, ddd, $J = 1.2$, 2.8, 50.4 Hz), 4.58 (1H, ddd, $J = 1.6$, 2.8, 17.6 Hz), 4.90–5.00 (2H, m), 5.72–5.83 (1H, m), 7.23–7.33 (5H, m) ppm; ^{13}C NMR (CDCl_3) 27.23, 28.65, 28.86, 32.46, 32.49, 48.55 (d, $J = 25.7$ Hz), 90.01 (d, $J = 20.6$ Hz), 114.27, 126.87, 127.80, 128.48, 138.94, 140.89, 168.09 (d, $J = 259.5$ Hz) ppm; ^{19}F NMR (CDCl_3) –99.46 (1F, ddd, $J = 17.6$, 17.6, 50.4 Hz) ppm; IR (neat) 3064, 3030, 2977, 2931, 2858, 1667, 1640, 1495, 1454, 1260, 1184, 994, 912, 849, 748, 699 cm^{-1} ; HRMS (EI) Calcd for $\text{C}_{16}\text{H}_{21}\text{F}$ (M^+) 232.1627, found 232.1635.

4.5.6. 2-Fluoro-6,6-(ethylenedioxy)-3-phenyl-1-hexene (**13f**)

Yield: 95%; ^1H NMR (CDCl_3) 1.55–1.73 (2H, m), 1.80–1.91 (1H, m), 2.01–2.10 (1H, m), 3.42 (1H, dt, $J = 16.0$, 8.0 Hz), 3.79–3.88 (2H, m), 3.90–3.98 (2H, m), 4.38 (1H, dd, $J = 2.8$, 50.4 Hz), 4.66 (1H, dd, $J = 2.8$, 17.2 Hz), 4.86 (1H, t, $J = 4.4$ Hz), 7.17–7.33 (5H, m) ppm; ^{13}C NMR (CDCl_3) 26.66 (d, $J = 4.1$ Hz), 31.61, 48.28 (d, $J = 25.6$ Hz), 64.88, 90.42 (d, $J = 20.6$ Hz), 104.12, 127.01, 127.85, 128.55, 140.34, 167.63 (d, $J = 260.34$ Hz) ppm; ^{19}F NMR (CDCl_3) –99.45 (1F, ddd, $J = 16.0$, 17.2, 50.4 Hz) ppm; IR (neat) 2957, 2882, 1669, 1495, 1454, 1411, 1256, 1179, 1141, 1032, 943, 911, 855, 701, 473 cm^{-1} ; HRMS (FAB) Calcd for $\text{C}_{14}\text{H}_{15}\text{FO}_2$ ($\text{M}-\text{H}$) 235.1135, found 235.1143.

4.5.7. 2-Fluoro-3-phenyl-1-butene (**13g**)

Yield: 12%; ^1H NMR (CDCl_3) 1.44 (3H, d, $J = 7.0$ Hz), 3.60 (1H, dq, $J = 14.0$, 7.0 Hz), 4.35 (1H, dd, $J = 2.8$, 50.4 Hz), 4.62 (1H, dd, $J = 2.8$, 17.6 Hz), 7.18–7.35 (5H, m) ppm; ^{13}C NMR (CDCl_3) 18.93 (d, $J = 4.9$ Hz), 42.36 (d, $J = 26.5$ Hz), 89.71 (d, $J = 20.7$ Hz), 126.88, 127.36, 128.53, 141.89, 169.13 (d, $J = 258.6$ Hz) ppm; ^{19}F NMR (CDCl_3) –97.44 (1F, ddd, $J = 14.0$, 17.6, 50.4 Hz) ppm; IR (neat) 2925, 2854, 1740, 1607, 1516, 1465, 1364, 1260, 1107, 804, 489 cm^{-1} .

4.5.8. 2-Fluoro-4-methyl-3-phenyl-1-pentene (**13h**)

Yield: 54%; ^1H NMR (CDCl_3) 0.74 (3H, d, $J = 6.7$ Hz), 1.07 (3H, d, $J = 6.6$ Hz), 2.18–2.23 (1H, m), 2.94 (1H, dd, $J = 27.4$, 10.3 Hz), 4.36 (1H, dd, $J = 2.8$, 50.0 Hz), 4.53 (1H, ddd, $J = 0.5$, 2.8, 17.6 Hz), 7.21–7.31 (5H, m) ppm; ^{13}C NMR (CDCl_3) 20.88, 21.38, 29.67, 57.32 (d, $J = 23.9$ Hz), 90.82 (d, $J = 20.8$ Hz), 126.81, 128.15, 128.40, 140.65, 167.09 (d, $J = 262.1$ Hz) ppm; ^{19}F NMR (CDCl_3) –103.01 (ddd, $J = 17.6$, 27.4, 50.0 Hz) ppm; IR (neat) 3063, 3029, 2961, 2931, 2872, 1666, 1494, 1467, 1387, 1250, 1193, 932, 849, 755, 700 cm^{-1} ; HRMS (EI) Calcd for $\text{C}_{12}\text{H}_{15}\text{F}$ (M^+) 178.1158, found 178.1151.

4.5.9. 3-Cyclohexyl-2-fluoro-3-phenyl-1-propene (**13i**)

Yield: 93%; ^1H NMR (CDCl_3) δ 0.74–2.01 (11H, m), 3.02 (1H, dd, $J = 10.4$, 28.4 Hz), 4.35 (1H, dd, $J = 2.7$, 50.0 Hz), 4.53 (1H, dd, $J = 2.7$, 17.6 Hz), 7.22–7.32 (5H, m) ppm; ^{13}C NMR (CDCl_3) δ 26.14, 26.37, 26.89, 30.16, 31.17, 31.77, 56.15 (d, $J = 23.7$ Hz), 90.95 (d, $J = 20.8$ Hz), 126.77, 128.22, 128.40, 140.32 (d, $J = 1.9$ Hz), 166.66 (d, $J = 262.5$ Hz) ppm; ^{19}F NMR (CDCl_3) δ –103.19 (1F, ddd, $J = 17.6$, 28.4, 50.0 Hz) ppm; IR (neat) 3062, 3028, 2926, 2852, 1666, 1495, 1250, 1179, 847, 754, 699 cm^{-1} ; HRMS (EI) Calcd for $\text{C}_{15}\text{H}_{19}\text{F}$ (M^+) 218.1471, found 218.1472.

4.5.10. 2-Fluoro-3-(4-methoxyphenyl)-1-heptene (**13m**)

Yield: 96%; ^1H NMR (CDCl_3) δ 0.89 (3H, t, $J = 7.0$ Hz), 1.22–1.35 (4H, m), 1.69–1.72 (1H, m), 1.89–1.92 (1H, m), 3.33 (1H, dt, $J = 17.7$, 7.7 Hz), 4.33 (1H, dd, $J = 2.8$, 50.2 Hz), 4.57 (1H, dd, $J = 2.8$, 17.7 Hz),

6.83–6.91 (2H, m), 7.17–7.20 (2H, m) ppm; ^{13}C NMR (CDCl_3) 13.91, 22.49, 29.59, 32.31 (d, $J = 3.4$ Hz), 47.70 (d, $J = 25.2$ Hz), 55.20, 89.73 (d, $J = 20.9$ Hz), 113.85, 128.77, 132.98, 158.47, 168.50 (d, $J = 260.0$ Hz) ppm; ^{19}F NMR (CDCl_3) –99.68 (1F, ddd, $J = 17.7, 17.7, 50.2$ Hz) ppm; IR (neat) 3033, 2999, 2958, 2862, 2837, 1881, 1667, 1611, 1513, 1465, 1251, 1178, 1038, 925, 828 cm^{-1} ; HRMS (FAB) Calcd for $\text{C}_{12}\text{H}_{19}\text{FO}$ (M^+) 222.1420, found 222.1423.

4.5.11. 6-(1-Fluoro-1-ethenyl)-1-dodecene (**13n**)

Yield: 89%; ^1H NMR (CDCl_3) δ 0.88 (3H, t, $J = 6.4$ Hz), 1.27–1.54 (14H, m), 2.02–2.15 (3H, m), 4.18 (1H, dd, $J = 2.4, 50.8$ Hz), 4.51 (1H, dd, $J = 2.4, 18.0$ Hz), 4.90–5.05 (2H, m), 5.79 (1H, ddt, $J = 16.8, 13.2, 6.8$ Hz) ppm; ^{13}C NMR (CDCl_3) 14.07, 22.64, 26.42, 27.09, 29.26, 31.42, 31.79, 31.98, 33.69, 42.62 (d, $J = 25.6$ Hz), 90.17 (d, $J = 20.7$ Hz), 114.48, 138.70, 168.26 (d, $J = 260.3$ Hz) ppm; ^{19}F NMR (CDCl_3) –106.98 (ddd, $J = 18.0, 18.0, 50.8$ Hz) ppm; IR (neat) 2930, 2859, 1667, 1642, 1460, 1257, 990, 912, 844, 458 cm^{-1} ; HRMS (FAB) Calcd for $\text{C}_{14}\text{H}_{25}\text{F}$ (M^+) 212.1940, found 212.1944.

4.5.12. (2S*, 3S*)-2-Benzoyloxy-3-(1-fluoroethen-1-yl)heptane (**13o**)

Yield: 89%; ^1H NMR (CDCl_3) δ 0.89 (3H, t, $J = 6.6$ Hz), 1.20 (3H, d, $J = 6.4$ Hz), 1.20–1.50 (6H, m), 2.33 (1H, ddt, $J = 25.6, 11.6, 6.0$ Hz), 3.61 (1H, dq, $J = 11.6, 6.4$ Hz), 4.31 (1H, ddd, $J = 51.2, 2.0, 0.8$ Hz), 4.48 (1H, d, $J = 11.6$ Hz), 4.58 (1H, d, $J = 11.6$ Hz), 4.65 (1H, dd, $J = 18.4, 2.0$ Hz), 7.24–7.37 (5H, m) ppm; ^{13}C NMR (CDCl_3) δ 13.96, 16.89, 22.65, 26.43 (d, $J = 2.1$ Hz), 29.55, 48.54 (d, $J = 30.1$ Hz), 71.06, 75.44, 91.91 (d, $J = 25.9$ Hz), 127.40, 127.58, 128.26, 138.77, 166.30 (d, $J = 260.3$ Hz) ppm; ^{19}F NMR (CDCl_3) δ –102.60 (1F, ddd, $J = 18.2, 25.6, 51.2$ Hz) ppm; IR (neat) 2957, 2871, 1669, 1496, 1455, 1379, 1343, 1259, 1147, 1109, 1028, 923, 847, 735, 697 cm^{-1} ; HRMS (FAB) Calcd for $\text{C}_{16}\text{H}_{24}\text{FO}$ ($\text{M} + \text{H}$) 251.1811, found 251.1807.

4.5.13. 2-Fluoro-4-methyl-3-phenyl-1-hexene (**13q**)

Yield: 90%; Diastereomeric ratio = 64:36; IR (neat) 3087, 3063, 3029, 2965, 2931, 2876, 1666, 1495, 1454, 1380, 1250, 848, 754, 699 cm^{-1} ; HRMS (EI) Calcd for $\text{C}_{13}\text{H}_{17}\text{F}$ (M^+) 192.1314, found 192.1301.

Major isomer: ^1H NMR (CDCl_3) 0.80 (3H, t, $J = 6.8$ Hz), 1.04 (3H, d, $J = 6.8$ Hz), 0.80–1.80 (3H, m), 3.00–3.15 (1H, m), 4.25–4.45 (1H, m), 4.48–4.55 (1H, m), 7.20–7.34 (5H, m); ^{19}F NMR (CDCl_3) δ –103.52 (1F, ddd, $J = 16.9, 29.3, 46.6$ Hz) ppm;

Minor isomer: ^1H NMR (CDCl_3) 0.70 (3H, d, $J = 6.4$ Hz), 0.94 (3H, t, $J = 7.2$ Hz), 0.80–1.80 (3H, m), 3.00–3.15 (1H, m), 4.25–4.45 (1H, m), 4.50–4.58 (1H, m), 7.20–7.34 (5H, m); ^{19}F NMR (CDCl_3) δ –102.72 (1F, ddd, $J = 17.0, 26.7, 51.1$ Hz) ppm.

4.5.14. (Z)-2-Fluoro-4,4-dimethyl-1-phenyl-1-pentene (**14r**)

Yield: 43%; Diastereomeric mixture: α -product/ γ -product = 88:12; IR (neat) 3060, 3027, 2957, 2905, 2871, 1685, 1366, 1238, 1137, 866, 754, 693 cm^{-1} ; HRMS (EI) Calcd for $\text{C}_{13}\text{H}_{17}\text{F}$ (M^+) 192.1317, found 192.1314.

(α -Product): ^1H NMR (CDCl_3) 1.02 (9H, s), 2.19 (2H, d, $J = 24.8$ Hz), 5.43 (1H, d, $J = 39.2$ Hz), 7.18–7.48 (5H, m) ppm; ^{19}F NMR (CDCl_3) δ –94.34 (1F, dt, $J = 24.8, 39.2$ Hz) ppm;

(γ -Product): ^{19}F NMR (CDCl_3) δ –97.28 (1F, ddd, $J = 17.3, 34.2, 51.6$ Hz) ppm.

4.5.15. 2-Fluoro-3-phenyl-1-pentene (**13u**)

Yield: 32%; ^1H NMR (CDCl_3) 0.91 (3H, t, $J = 7.2$ Hz), 1.69–1.80 (1H, m), 1.91–2.01 (1H, m), 3.28 (1H, dt, $J = 17.6, 8.0$ Hz), 4.36 (1H, dd, $J = 50.0, 2.8$ Hz), 4.61 (1H, ddd, $J = 17.2, 2.8, 0.8$ Hz), 7.23–7.34 (5H, m) ppm; ^{13}C NMR (CDCl_3) δ 12.12, 25.64 (d, $J = 3.3$ Hz), 50.35 (d, $J = 25.6$ Hz), 90.23 (d, $J = 20.7$ Hz), 126.88, 127.86, 128.46, 140.73, 167.91 (d, $J = 259.5$ Hz) ppm; ^{19}F NMR (CDCl_3) δ –99.55 (1F, dd, $J = 17.2, 17.6, 50.0$ Hz) ppm; IR (neat) 3063, 3030, 2967,

2934, 2877, 1668, 1495, 1454, 1381, 1261, 1184, 1073, 929, 851, 764, 699 cm^{-1} ; HRMS (EI) Calcd for $\text{C}_{11}\text{H}_{13}\text{F}$ (M^+) 164.1001, found 164.0995.

4.6. Determination of the stereochemistry of **13o**

A mixture of ozone and oxygen was gently bubbled through a stirred solution of **13o** (0.075 g, 0.30 mmol) in CH_2Cl_2 (4 mL) at -78°C until the color of the solution became pale blue. The excess ozone was purged from the solution by bubbling oxygen for 5 min. Dimethyl sulfide (0.06 mL) was added and the reaction mixture was allowed to warm to room temperature over 1 h. The resulting mixture was concentrated *in vacuo*.

To a suspension of lithium aluminum hydride (an excess amount) in THF (2 mL) was added a solution of the above residue in THF (2 mL) at 0°C . The whole was stirred for 15 min at that temperature, and then the reaction mixture was poured into sat. NH_4Cl aq. The whole was extracted with EtOAc three times. The combined organic layers were dried over anhydrous Na_2SO_4 , filtered and evaporated under the reduce pressure. Purification of the residue by silica gel column chromatography gave the corresponding alcohol **15** (0.04 g, 0.17 mmol, 58% yield).

To a solution of the alcohol **15** (40 mg, 0.17 mmol) in EtOH (2 mL) was added 10% Pd/C (18 mg, 10 mol%) under Ar atmosphere. The argon was replaced by H_2 gas, and the mixture was stirred overnight at room temperature. The resulting mixture was filtered with the celite pad and the filtrate was concentrated *in vacuo*. The crude sample was purified by silica gel column chromatography to afford the corresponding diol **16** (0.012 g, 0.085 mmol, 50% yield).

2,2-Dimethoxypropane (0.15 g, 1.5 mmol) and TsOH- H_2O (0.001 g, 10 mol%) were successively added to a solution of the diol **16** (12 mg, 0.085 mmol) in DMF (1.5 mL) at room temperature. After being stirred for 2 days, the resulting mixture was subjected to a usual aqueous work up. Purification of the crude product by silica gel column chromatography gave the corresponding (4S*,5R*)-5-*n*-butyl-2,2,4-trimethyl-1,3-dioxane (**17**) (0.003 g, 24%). Yield: 24%; ^1H NMR (CDCl_3) 0.89 (3H, t, $J = 6.0$ Hz), 1.18 (3F, d, $J = 5.6$ Hz), 1.18–1.39 (7H, m), 1.39 (3H, s), 1.43 (3H, s), 3.54 (1H, dd, $J = 11.6, 11.6$ Hz), 3.65 (1H, dq, $J = 11.6, 5.6$ Hz), 3.81 (1H, dd, $J = 4.8, 11.6$ Hz) ppm; ^{13}C NMR (CDCl_3) 13.91, 19.21, 19.78, 22.94, 27.97, 28.58, 29.78, 40.88, 64.45, 70.11, 97.80 ppm.

4.7. Ring-closing metathesis

To a solution of 2-fluoro-3-phenyl-1,7-heptadiene (0.031 g, 0.15 mmol) in CH_2Cl_2 (15 mL) was added ruthenium complex (0.026 g, 20 mol%). The mixture was heated in a sealed vial for 2 h at 100°C , then cooled again. After concentration of the reaction mixture, the resulting brown oil was purified by column chromatography to give 1-fluoro-6-phenyl-cyclohexene (**18c**) (0.010 g, 38% yield).

4.7.1. 1-Fluoro-6-phenylcyclohexene (**18c**)

Yield: 38%; ^1H NMR (CDCl_3) 1.51–1.58 (2H, m), 1.71–1.80 (1H, m), 2.05–2.21 (3H, m), 3.62 (1H, m), 5.48 (1H, dt, $J = 17.6, 4.0$ Hz), 7.21–7.38 (5H, m) ppm; ^{13}C NMR (CDCl_3) 19.14 (d, $J = 2.0$ Hz), 23.11 (d, $J = 7.5$ Hz), 32.96 (d, $J = 6.6$ Hz), 42.51 (d, $J = 23.0$ Hz), 104.90 (d, $J = 14.9$ Hz), 126.56, 127.85, 128.39, 141.98 (d, $J = 2.5$ Hz), 159.46 (d, $J = 254.6$ Hz) ppm; ^{19}F NMR (CDCl_3) –105.56 (d, $J = 17.6$ Hz) ppm; HRMS (FAB) Calcd for $\text{C}_{12}\text{H}_{13}\text{F}$ (M^+) 176.1001, found 176.0994.

4.7.2. 2-Fluoro-3-*n*-hexylcyclohexene (**18n**)

Yield: 100%; ^1H NMR (CDCl_3) 0.80 (3H, t, $J = 6.0$ Hz), 1.28–1.39 (10H, m), 1.39–1.83 (4H, m), 1.98–2.04 (2H, m), 2.27 (1H, m), 5.15 (1H, dt, $J = 18.4, 3.6$ Hz) ppm; ^{13}C NMR (CDCl_3) 14.08, 20.04, 20.05,

22.66, 23.04 (d, $J = 7.4$ Hz), 26.91, 28.11, 28.18, 31.84 (d, $J = 1.7$ Hz), 35.71 (d, $J = 21.5$ Hz), 102.00 (d, $J = 16.6$ Hz), 162.79 (d, $J = 254.6$ Hz) ppm; ^{19}F NMR (CDCl_3) -109.29 (d, $J = 18.4$ Hz) ppm; IR (neat) 2929, 2859, 1697, 1458, 1377, 1337, 1281, 1134, 865, 812, 723 cm^{-1} ; HRMS (EI) Calcd for $\text{C}_{12}\text{H}_{21}\text{F}$ (M^+) 184.1627, found 184.1632.

4.8. Preparation of **24**

4.8.1. Deprotection of the acetal **13f**

To a mixture of THF/1 M hydrochloric acid (20 mL/20 mL) was added 2-fluoro-3-phenyl-6,6-(ethylenedioxy)-1-hexene (**13f**, 0.24 g, 1.02 mmol) at room temperature. After stirring of the reaction mixture for 24 h, the solution was extracted with diethyl ether three times, and the combined organic layers were washed with an aqueous saturated NaHCO_3 solution, dried over anhydrous Na_2SO_4 , concentrated *in vacuo*. The residue was purified by silica gel column chromatography to give 5-fluoro-4-phenyl-5-hexenal (**20**) (0.13 g, 0.67 mmol, 66% yield). Yield: 66%; ^1H NMR (CDCl_3) 2.02–2.11 (1H, m), 2.25 (1H, m), 2.39–2.56 (2H, m), 3.44 (1H, ddd, $J = 8.0, 8.0, 17.6$ Hz), 4.40 (1H, dd, $J = 2.8, 50.0$ Hz), 4.65 (1H, dd, $J = 2.8, 17.6$ Hz), 7.20–7.40 (5H, m) ppm; ^{13}C NMR (CDCl_3) 24.76 (d, $J = 4.1$ Hz), 41.62, 47.53 (d, $J = 25.7$ Hz), 90.95 (d, $J = 19.8$ Hz), 127.35, 127.80, 128.73, 139.60, 166.98 (d, $J = 260.3$ Hz), 201.33 ppm; ^{19}F NMR (CDCl_3) -99.95 (1F, ddd, $J = 17.6, 17.6, 50.0$ Hz) ppm; IR (neat) 3062, 3030, 2940, 2890, 2827, 2727, 1724, 1670, 1602, 1495, 1454, 1390, 1256, 1183, 1065, 919, 857, 753, 701 cm^{-1} ; HRMS (EI) Calcd for $\text{C}_{12}\text{H}_{13}\text{FO}$ (M^+) 192.0950, found 192.0957.

4.8.2. Reaction of the aldehyde **20** with vinyl magnesium bromide

To the aldehyde (**20**, 0.13 g, 0.67 mmol) in dry Et_2O (5 mL) was added dropwise a solution of vinyl magnesium bromide (1 M in THF, 0.81 mL, 0.81 mmol) at room temperature. After stirring for 1 h, the reaction was quenched with cold water and the whole was extracted with diethyl ether. The combined organic layers were dried over anhydrous Na_2SO_4 , concentrated *in vacuo*. The residue was purified by silica gel column chromatography to give 7-fluoro-6-phenyl-1,7-octadiene-3-ol (**21**) (0.12 g, 0.53 mmol, 79%). Diastereomeric ratio = ca. 1:1; Yield: 79%; ^1H NMR (CDCl_3) 1.42–1.50 (1H, m), 1.52–1.63 (2H, m), 1.72–2.09 (2H, m), 3.40 (1H, ddd, $J = 8.0, 8.0, 17.6$ Hz), 4.07–4.14 (1H, m), 4.43 (1H, dd, $J = 2.8, 50.4$ Hz), 4.61 (1H, dd, $J = 2.8, 17.6$ Hz), 5.10 (1H, m), 5.21 (1H, m), 5.77–5.88 (1H, m), 7.23–7.34 (5H, m) ppm; ^{19}F NMR (CDCl_3) -99.70 to -99.40 (1F, m) ppm; IR (neat) 3362, 3063, 3029, 2935, 2871, 1669, 1495, 1454, 1256, 1182, 927, 853, 700 cm^{-1} ; HRMS (FAB) Calcd for $\text{C}_{14}\text{H}_{17}\text{FNaO}$ ($\text{M} + \text{Na}$) 243.1161, found 243.1151.

4.8.3. Reaction of **21** in the presence of Grubbs 2nd catalyst

To a solution of **21** (0.033 g, 0.15 mmol) in CH_2Cl_2 (15 mL) was added ruthenium complex (0.026 g, 20 mol%). The mixture was heated in a sealed vial for 2 h at 100 °C, then cooled again. After concentration of the reaction mixture, the resulting brown oil was purified by silica gel column chromatography to give 7-fluoro-6-phenyl-7-octen-3-one (**22**) (0.029 g, 88% yield). ^1H NMR (CDCl_3) 1.01 (3H, dt, $J = 7.6, 1.2$ Hz), 1.99–2.07 (1H, m), 2.07–2.23 (1H, m), 2.32–2.45 (4H, m), 3.42 (1H, dt, $J = 17.2, 8.4$ Hz), 4.38 (1H, dd, $J = 50.4, 3.2$ Hz), 4.62 (1H, dd, $J = 18.0, 3.2, 0.8$ Hz), 7.23–7.36 (5H, m) ppm; ^{13}C NMR (CDCl_3) 7.76, 26.26 (d, $J = 4.1$ Hz), 35.98, 39.65, 47.51 (d, $J = 25.6$ Hz), 90.56 (d, $J = 20.7$ Hz), 127.15, 127.81, 128.62, 139.98, 167.33 (d, $J = 260.3$ Hz), 210.68 ppm; ^{19}F NMR (CDCl_3) -99.53 (1F, ddd, $J = 17.2, 18.0, 50.4$ Hz) ppm; IR (neat) 3029, 2976, 2939, 1714, 1670, 1496, 1454, 1374, 1260, 1115, 855, 700, 482 cm^{-1} ; HRMS (FAB) Calcd for $\text{C}_{14}\text{H}_{18}\text{FO}$ ($\text{M} + \text{H}$) 221.1342, found 221.1328.

4.8.4. Preparation of methyl ether

To a suspension of NaH (8 mg, 55% included, 0.64 mmol) in THF (2 mL) was added a THF solution of **21** (0.12 g, 0.53 mmol) and MeI (0.083 mL, 1.33 mmol) in this order at 0 °C. The reaction was allowed to warm to room temperature, and the whole was stirred overnight. Then the reaction was poured into sat. NH_4Cl aq. and then the whole was extracted with EtOAc three times. The combined organic layers were dried over anhydrous Na_2SO_4 , filtered and concentrated *in vacuo*. The residue was purified by silica gel column chromatography to give 2-fluoro-6-methoxy-3-phenyl-1,7-octadiene (**23**) (0.11 g, 0.46 mmol, 86%).

Diastereomeric ratio = ca. 1:1; Yield: 86%; ^1H NMR (CDCl_3) 1.37–2.05 (4H, m), 3.24 or 3.25 (3H, s), 3.36 (1H, dt, $J = 16.0, 6.8$ Hz), 3.49 (1H, m), 4.37 (1H, dd, $J = 2.8, 50.0$ Hz), 4.60 (1H, dd, $J = 2.8, 18.0$ Hz), 5.15–5.21 (2H, m), 5.55–5.66 (1H, m), 7.18–7.33 (5H, m) ppm; ^{19}F NMR (CDCl_3) δ -99.75 (1F, ddd, $J = 18.0, 18.0, 50.0$ Hz) or -99.45 (1F, ddd, $J = 18.0, 18.0, 50.0$ Hz) ppm; IR (neat) 3063, 3029, 2980, 2935, 2821, 1669, 1495, 1454, 1421, 1257, 1183, 1104, 927, 852, 700 cm^{-1} .

4.8.5. Ring-closing metathesis of **23**

To a solution of **23** (0.035 g, 0.15 mmol) in CH_2Cl_2 (15 mL) was added ruthenium complex (0.026 g, 20 mol%). The mixture was heated in a sealed vial for 2 h at 100 °C, and then cooled again. After concentration of the reaction mixture, the resulting brown oil was purified by silica gel column chromatography to give 1-fluoro-3-methoxy-6-phenyl-1-cyclohexene (**24**) (0.023 g, 73% yield). Diastereomeric ratio = ca. 7:3; ^1H NMR (CDCl_3) δ 1.60–2.10 (4H, m), 3.38 (3H, s) or 3.39 (3H, s), 3.40–3.65 (1H, m), 3.95–4.05 (1H, m), 5.60–5.71 (1H, m), 7.20–7.35 (5H, m) ppm; ^{19}F NMR (CDCl_3) -101.94 (1F, d, $J = 17.3$ Hz) or -109.19 (1F, d, $J = 14.7$ Hz) ppm; IR (neat) 3063, 3029, 2932, 2868, 2820, 1693, 1603, 1495, 1454, 1370, 1192, 1139, 1089, 893, 855, 755, 701 cm^{-1} ; HRMS (FAB) Calcd for $\text{C}_{13}\text{H}_{15}\text{FO}$ (M^+) 206.1107, found 206.1111.

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